

The Influence of Health Cadre Involvement on the Quality of Public Health Services in the Puskesmas Working area

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ABSTRACT

Health cadres play a strategic role as a bridge between communities and primary health care facilities, particularly in promotive and preventive activities. Limited cadre involvement may contribute to suboptimal health service quality, including low Posyandu coverage, delayed detection of health problems, and limited community participation. This study aimed to analyze the relationship between health cadre involvement and public health service quality based on patient perceptions in the working area of Ambacang Padang Health Center. A quantitative cross-sectional design was employed involving patients who received health services at the health center. A total of 96 respondents were selected using accidental sampling based on predetermined inclusion criteria. Data were collected using validated and reliable structured questionnaires assessing cadre involvement and perceived health service quality. Data were analyzed using univariate analysis and the Chi-square test at $\alpha=0.05$. Results showed that 54.2% of patients perceived cadre involvement as low, while 57.3% perceived health service quality as poor. A significant relationship was found between cadre involvement and health service quality ($p=0.000$; OR=3.98; 95% CI: 1.76–8.99). Strengthening cadre training, supervision, and performance monitoring is recommended to enhance primary health service quality and community participation.

Keywords: Health Cadres, Service Quality, Patients, Puskesmas

INTRODUCTION

Public service is a crucial aspect in the organization of modern organizations, including in the field of health, because in it there is an interaction between service providers, service recipients, and the types of services themselves. Good public service capacity is highly dependent on teamwork, inter-organizational relationships, professional distribution of tasks, community involvement, positive work culture, and the support of organizational leaders. Public service agencies in the field of health such as health centers are required to have a strong commitment to the principles of good governance so that the quality of service can be maintained sustainably. Ideally, people want the best health services, but the gap between regulation and reality on the ground is still a recurring issue and makes the topic of health care quality relevant to be studied until now. (Tua F S & Syofian, 2020)

Pusat Kesehatan Masyarakat (Puskesmas) is a first-level health service facility that organizes individual and public health efforts by prioritizing promotive and preventive efforts in its work area. Puskesmas serves as a functional service unit that plays a role in health development, a community participation development center, as well as a first strata Health Service Center for residents in their work area. The quality of service in puskesmas has a very important role in ensuring that people get the right care according to their needs. Because of its position as the frontline of the health system, the quality of service at the puskesmas level will largely determine the success of overall health development before cases are referred to higher health facilities. (Dohona, Sometimes, Hutabarat, & Tumewu, 2023)

Nationally, the existence of puskesmas as the spearhead of Primary Health Services continues to grow as people's needs for equitable access to health services increase. Based on Indonesia's health profile, the number of health centers throughout Indonesia has reached more than ten thousand units spread across various provinces, with variations in the number of inpatient and non-inpatient health centers in each region. The scale of such a large facility requires consistent service quality standards so that there is no quality gap between regions. With its strategic position, puskesmas is required to be able to integrate individual, family, and community services as a whole in order to strengthen public health determinants at the basic level. (Ministry of Health, 2024)

The same context also applies in the city of Padang, West Sumatra province, which has dozens of puskesmas units as basic health care providers for its residents. Each Health Center in Padang city is generally equipped with medical personnel such as general practitioners and dentists with numbers adjusted to population density and volume of patient visits in their respective work areas. The availability of adequate health workers is one of the important prerequisites so that services at the puskesmas level can run optimally and be responsive to the needs of the community. However, the number of professionals available will not be enough to reach all levels of society without the support of the elements of society itself, one of which is through the role of health cadres at the community level. (Department Of Health Of Padang City, 2024)



The involvement of community elements in the implementation of Health Services is one of the tangible manifestations of the National Health System that is oriented to community empowerment. The government through various policies encourages the creation of health services that not only rely on professionals, but also involve the active participation of the community as partners in promotional and preventive efforts. The differences that continue to emerge between the ideal expectations of National Health Services and the reality of their implementation in the field make the issue of community involvement, including health cadres, a topic that is always interesting to be studied more deeply. Health cadres are ultimately positioned as a strategic bridge between the formal service system and the real needs of the community at the grassroots. (Giri, 2021)

Health cadres are members of the community who are selected and come from the local community to assist in the implementation of various health activities, especially at the posyandu level and puskesmas work areas. Given the large role of cadres in the management of community-based health activities, increasing the competence of cadres through training and continuing education is a necessity that cannot be ignored. Skilled and well-informed cadres will be able to produce more reliable performance in supporting various health programs in their region. Without adequate guidance, the potential of cadres as the spearhead of community-based services is at risk of not being optimally utilized. (Nurbaya, Haji Saeni, & Irwan, 2022)

Functionally, health cadres have a significant role in various aspects of public health services, ranging from early detection of diseases, collection of citizen health data, health education, implementation of health programs in the field, to referral of cases to higher health care facilities. This multidimensional role makes cadres one of the important elements that support the success of primary health programs at the Puskesmas work area level. Optimal cadre activity will have a direct impact on the speed of addressing health problems in the community before their condition worsens. Conversely, weak cadre involvement can slow down early detection and hinder the achievement of health program targets in a region. (J-KESMAS: Journal of Public Health, 2022)

Although the role of health cadres is very important, various studies show that the activeness and involvement of cadres in the field is still a problem that has not been fully resolved. Factors such as motivation, family support, incentives, and the double workload between domestic and social roles also affect the level of activity of cadres in carrying out their duties. This condition causes that not all cadres are able to contribute consistently in every health activity held in the working area of the puskesmas. As a result, variations in the level of cadre involvement between regions also resulted in significant variations in the achievement of public health programs. (Oruh, 2021)

On the other hand, Human Resource Management in puskesmas contributes greatly to the quality of services provided, either through proportional distribution of tasks, continuous training, or improving the work discipline of health workers and cadres involved in it. However, various human resource problems in the health care sector such as lack of adequate personnel and gaps in

the distribution of skilled workers are still often found in various areas of puskesmas work. These conditions ultimately have a significant impact on the quality of health workers and services provided to the community. Integrated coaching between health professionals and cadres is the key to closing the gap. (Hasibuan et al., 2024; Wati et al., 2023)

Community participation in community-based health activities, such as posyandu, is strongly influenced by the activeness and involvement of cadres in mobilizing citizens in the region. Cadres who are active in personal approach and socialization on a regular basis tend to be able to increase the number of attendance and community participation in health programs. Conversely, the lack of interaction between cadres and residents can reduce people's interest in utilizing health services available in the puskesmas work area. This shows that the involvement of cadres not only has an impact on the technical aspects of service, but also on the participative social aspects of society as a whole. (Sintiawati et al., 2021)

The management of the implementation of health monitoring activities carried out by cadres is also closely related to the level of community participation in health activities in the region. When cadres are able to manage the implementation of growth and health monitoring in an orderly and consistent manner, the community will be more confident and more routinely utilize the services provided. This relationship indicates that the quality of cadre involvement cannot be separated from the quality of health services that are directly felt by the community as service recipients. Thus, strengthening the role of cadres is one of the important strategies to improve the quality of primary health services in a sustainable manner. (Nisa et al., 2022)

In terms of service recipients, patient satisfaction is often used as a measure of the effectiveness and quality of health services provided by primary health facilities, including health centers. The quality of health services refers to the extent to which the services provided are able to meet the needs and expectations of patients, both in terms of reliability, responsiveness, assurance, empathy, and physical evidence of services. A quality health care system is one of the crucial aspects in meeting the needs of the community for health services available in the region. The patient's assessment of the quality of the service is also influenced by how far the supporting elements of the service, including health cadres, play a role in the service process they receive. (Africa, 2023)

A number of studies have shown that patient satisfaction levels have a significant impact on overall health outcomes, where patients who feel satisfied with the services received tend to be more obedient to medical instructions and have a better recovery process. Satisfied patients also tend to seek alternative health services outside of official facilities less often, so that continuity of care can be better maintained. These findings reinforce the importance of understanding the supporting factors that affect the satisfaction and quality of health services, including the involvement of cadres as part of the primary service support system. Therefore, the study of the role of cadres in the context of service quality is becoming increasingly relevant to continue to be developed. (Firaldi, 2023; Dewi Lukita Sari, 2020)



The urgency of the role of cadres is increasingly evident in specific health issues such as reducing stunting rates, where posyandu cadres contribute greatly through early detection of the nutritional status of toddlers, education to mothers, and regular growth monitoring. Optimal cadre involvement in these programs has been proven to contribute to the improvement of public health indicators in the puskesmas work area. Conversely, when cadre involvement is low, the achievement of priority health programs such as stunting prevention and toddler nutrition monitoring becomes less than optimal. This condition confirms that the involvement of cadres is not just complementary, but a core component that determines the success of health programs based on the working area of the puskesmas. (Hamdy et al., 2023)

Based on the series of descriptions, it appears that the involvement of health cadres is closely related to the quality of Public Health Services, both at the national level and in the context of specific puskesmas working areas, including in the Ambacang Padang Puskesmas. Preliminary studies in the Ambacang Puskesmas work area showed indications of variations in the involvement of health cadres between polyclinics, while community assessments of the quality of health services in the region also showed considerable differences. The gap between the ideal role of cadres as outlined in various previous studies with real conditions in the field is what encourages researchers to examine further on the influence of the involvement of health cadres on the quality of Public Health Services. Researchers are interested in looking at this issue from the perspective of patients as direct recipients of services, with the hope that the results can provide an empirical picture that can be used as a basis for the Ambacang Padang Health Center in formulating a more targeted cadre development policy for improving the quality of public health services in its work area.

METHODS

This study aims to analyze the influence of the involvement of health cadres on the quality of Public Health Services according to the assessment of patients in the work area of the Ambacang Padang Health Center. This type of research is a quantitative study with cross sectional design, where the independent variable (involvement of health cadres) and the dependent variable (quality of health services) are measured simultaneously at one time. The population in this study is all patients who visit and get services in the Working Area of Ambacang Padang Health Center. The number of samples was determined as many as 96 respondents calculated using the Slovin formula with a 90% confidence level, and selected using accidental sampling technique, which is sampling of patients who happened to be encountered at the research site at the time of data collection took place. The sample inclusion criteria include patients aged 17 years, registered as active patients in the Ambacang Padang Health Center working area, have received health services at least once in the last six months, domiciled in the puskesmas working area, and willing to be respondents to the study. The exclusion criteria include patients in emergency conditions, experiencing cognitive or communication disorders that impede filling out the questionnaire, as well as patients who are not willing to continue filling out the questionnaire until it is completed.

Data collection technique is done directly (primary data) through the filling of structured questionnaires that have been tested for validity using Pearson product moment correlation test and tested for reliability using Cronbach's Alpha with coefficient value >0.60 , so that the instrument is declared valid and reliable to use. The research instrument consists of three parts, namely the respondent characteristic data questionnaire, the health cadre involvement questionnaire prepared based on indicators of cadre roles in promotive and preventive activities, and the health service quality questionnaire prepared based on the dimensions of service quality (physical evidence, reliability, responsiveness, assurance, and empathy) using the Likert scale. Data analysis was conducted in univariate to see the frequency distribution of each variable, and bivariate using Chi-Square test to analyze the relationship between the involvement of health cadres with the quality of health services, with the significance level of $\alpha = 0.05$ (the relationship is declared meaningful if the value of $p < 0.05$). This study has taken into account the principles of research ethics with each respondent given a research explanation sheet and asked to sign a consent sheet (informed consent) before filling out the questionnaire, while upholding the principles of autonomy (the right to refuse or resign at any time without consequences), anonymity (confidentiality of the respondent's identity), and confidentiality (confidentiality of data that is only used for research purposes).

RESULTS

1. Respondent Characteristics

Based on the results of research conducted on 96 respondents in the Ambacang Padang Community Health Center work area, the following description of the characteristics of respondents was obtained based on age, gender, education, and occupation.

Table 1. Respondent Characteristics

No	Characteristics	Categories	Frequency (n)	Percentage (%)
1	Age	17-25 years old	18	18,8
		26-35 years old	27	28,1
		36-45 years old	25	26,0
		>45 years old	26	27,1
2	Gender	Men	34	35,4
		Girls	62	64,6
3	Education	Primary (SD-SMP)	31	32,3
		Secondary (SMA)	45	46,9
		Height (D3/S1)	20	20,8
4	Jobs	Works	53	55,2
		Not Working	43	44,8



No	Characteristics	Categories	Frequency (n)	Percentage (%)
	Total		96	100

The table above shows that the majority of respondents are in the productive age range of 26-35 years (28.1%), with the majority being female (64.6%). In terms of education, more than a third of respondents (46.9%) have secondary education (SMA), while in terms of employment, most respondents (55.2%) have working status. This characteristic illustrates that patients who become respondents are generally productive age groups with secondary education backgrounds, so they are representative enough to objectively assess the quality of health services they receive.

2. Variable Frequency Distribution of Health Cadre Involvement and Health Service Quality Based on Patient Assessment

Analisis univariat dilakukan untuk melihat distribusi frekuensi variabel keterlibatan kader kesehatan dan kualitas pelayanan kesehatan berdasarkan penilaian pasien.

Table 2. Variable Frequency Distribution of Health Cadre Involvement and Health Service Quality Based on Patient Assessment

No	Variable	Categories	Frequency (n)	Percentage (%)
1	Health Cadre Involvement	Low	52	54,2
		Height	44	45,8
2	Quality Of Health Services	Less Good	55	57,3
		Good	41	42,7
Total			96	100

The results of univariate analysis showed that more than half of respondents (54.2%) assess the involvement of health cadres in the Ambacang Padang Health Center working area is still in the low category. In line with this, most respondents (57.3%) also assess the quality of health services they receive is still not good. These findings indicate that the role of health cadres in supporting promotive and preventive activities in the region is not optimal, which is in line with patient assessment of the quality of health services that still need to be improved.

3. The Relationship Between the Involvement of Health Cadres With the Quality of Health Care According to Patient Assessment

Bivariate analysis was conducted to determine the relationship between the involvement of health cadres with the quality of Health Services according to the assessment of patients, using Chi-Square test with the level of significance $\alpha = 0.05$.

Table 3. The Relationship Between the Involvement of Health Cadres With the Quality of Health Care According to Patient Assessment

Health Cadre Involvement	Poor Quality Of Service n (%)	Good Service Quality n (%)	Total n (%)	p-value	OR (95% CI)
Low	38 (73,1)	14 (26,9)	52 (100)		
Height	17 (38,6)	27 (61,4)	44 (100)		
Total	55 (57,3)	41 (42,7)	96 (100)	0,000	4,31 (1,89–9,84)

The results of The Chi-Square test showed that of 52 respondents with low health cadre involvement, 73.1% rated the quality of health services as poor, while of 44 respondents with high cadre involvement, only 38.6% rated the quality of services as poor. Obtained value $p=0.000$ ($p<0.05$), which means there is a statistically significant relationship between the involvement of health cadres with the quality of Health Services according to the assessment of patients in the work area of the Health Center Ambacang Padang. The value of the Odds Ratio (OR) of 4.31 (95% CI: 1.89–9.84) indicates that patients who assess the involvement of low health cadres have a 4.31 times greater risk of assessing the quality of health care is not good than patients who assess the involvement of high health cadres. Thus, it can be concluded that the higher the involvement of health cadres, the better the quality of Health Services perceived by the community in the working area of the health center.

DISCUSSION

1. Variable Frequency Distribution of Health Cadre Involvement and Health Service Quality Based on Patient Assessment

The results of the univariate analysis in Table 2 showed that of the 96 respondents studied, more than half of the respondents, namely 52 people (54.2%) assessed the involvement of health cadres in the Ambacang Padang Health Center work area, were in the low category, while 44 people (45.8%) assessed the involvement of cadres in the high category. In the variable quality of health care, as many as 55 people (57.3%) respondents assess the quality of health care they receive is still not good, while 41 people (42.7%) assess the quality of health care is good. This distribution shows that the proportion of negative assessments on both variables is relatively more dominant than positive assessments. The findings in this frequency distribution table are the basis for further discussion about the picture of health cadre involvement and the quality of health services in the Ambacang Padang Health Center working area.

The high proportion of respondents who assess the involvement of health cadres is still low illustrates that the role of cadres in promotive and preventive activities in the working area of the health center is not fully optimal. This condition is in line with the theory that community health worker engagement is strongly influenced by knowledge, intrinsic motivation, workload, and supervision system support received by cadres from local health facilities. When one of these



factors is not adequately met, the intensity of cadre involvement in carrying out their duties in the community tends to decrease. The low involvement of cadres found in this study indicates that there is a gap between the ideal role of cadres as expected in the primary health system and the real conditions for its implementation in the field. (Gunawan & Ayubi, 2023)

The picture of low cadre involvement in more than half of respondents is also in line with the results of the systematic review which found that the workload of public health cadres in Indonesia is relatively high without adequate supervision and support mechanisms from first-rate health facilities. The double workload between the domestic role as a family member and the social role as a cadre is often a barrier that limits the time and energy of cadres to actively engage in various health activities. The condition is aggravated if cadres do not get sustainable development and adequate incentives as a form of appreciation for their contributions. This further reinforces the picture that the low involvement of cadres is not merely an individual issue, but also a reflection of the weak support system available to cadres at the Puskesmas work area level. (Nasrin, Sumayyah, Listiana, & Espressivo, 2024)

However, there were 44 respondents (45.8%) who considered the involvement of health cadres to be in the high category, which shows that some cadres in the Ambacang Padang Health Center work area are still able to carry out their roles actively despite being in a densely populated urban environment. Cadres with high involvement generally have strong motivation, both intrinsic motivation in the form of a sense of Social Responsibility and extrinsic motivation in the form of family and environmental support. High motivation is proven to be one of the main determinant factors that encourage cadres to remain active in contributing to health activities despite facing various resource limitations. The proportion of cadres with high involvement is an important social capital and needs to be maintained and further developed by the puskesmas. (Manoppo & Rumambi, 2022)

The low involvement of cadres in this study is also relevant to the results of research in the working area of the Padang City Water children's Health Center which found that factors such as age, education, knowledge, and motivation were significantly related to the performance of cadres in the implementation of posyandu activities. The similarity of the context of the same city area, namely Padang City, makes the findings relevant as a direct comparison to the results of this study. The consistency of the pattern of low cadre involvement in two different puskesmas working areas in the same city indicates that this problem is quite systemic and does not just happen by chance at one research site. Therefore, a thorough evaluation of the cadre development system at the Padang city level needs to be the concern of the local health office. (Raniwati, Ernawati, Sari, Sari, & Astuti, 2022)

In terms of community nursing theory, the involvement of health cadres can be understood as a form of community participation in community-based health efforts that involve a process of empowerment (empowerment) on an ongoing basis. This empowerment process ideally runs through the stages of education, assistance, and independence of cadres in carrying out their functions without always relying on the direction of professional health workers. If the

empowerment process is stopped at the early education stage without ongoing assistance, the involvement of cadres tends to stagnate or even decrease over time. The Data in this study which shows the dominance of low involvement category indicates that the process of empowerment of cadres in the Ambacang Padang Health Center work area may not be running optimally and sustainably. (Sukmawati, Hermayanti, Fadlyana, Maulana, & Mediani, 2025)

Furthermore, in the variable of Health Care Quality, the proportion of respondents who rated the service as not good (57.3%) was higher than those who rated it good (42.7%), which indicates that the quality of health care in the Ambacang Padang Puskesmas working area according to patient perceptions still needs to be improved. Based on the theory of health care quality dimensions proposed by Parasuraman, patient assessment of service quality is formed from five main dimensions, namely physical evidence, reliability, responsiveness, assurance, and empathy. When one or more of these dimensions, especially the dimensions of responsiveness and empathy that are closely related to interpersonal interaction, are not met to the fullest, the patient's assessment of the overall quality of service tends to be less good. The predominant proportion of negative assessments in this study indicates a gap between the expectations of patients and the reality of the services they receive in primary health facilities. (Jonkisz, 2021)

The findings on the dominance of poor service quality assessment is in line with the results of research in Bailang Puskesmas Manado city, which shows that the quality of health services in puskesmas still faces various challenges, especially on the dimensions of the responsiveness of officers and the speed of the Administrative Service process. Similar challenges are often found in health centers with a high volume of patient visits, where the ratio between the number of health workers and the number of patients served is not fully proportional. This condition causes the service time to be longer and the interaction between staff and patients to be less than optimal, resulting in an impact on the patient's assessment of the overall quality of Service. The similarity of the pattern of findings at various locations of this study reinforces the picture that the issue of service quality in health centers is a fairly common issue found in primary health facilities in Indonesia. (Maameah, Rumayar, & Mandagi, 2022)

On the other hand, there are 41 respondents (42.7%) who assess the quality of Health Services is good, which indicates that some patients still feel a positive service experience at the Health Center Ambacang Padang. This group of respondents is most likely a patient who has long been a user of puskesmas services and has a fairly good interpersonal relationship with health workers and cadres who accompany them. In addition, the factor of conformity between patient expectations and the reality of the services received also influenced the positive assessment, considering that patient expectations of first-level health facilities are generally not as high as expectations of referral health facilities. This proportion of good assessment remains a positive capital that can be used as a reference for puskesmas in identifying service practices that have been running well to be replicated more widely. (Nurmawati & Pramesti, 2022)

The frequency distribution pattern of health service quality in this study is also relevant to the results of research on the relationship of service quality with the interest in re-utilization of



health services in the Health Center Tamalanrea Makassar city, which found that patient perceptions of service quality largely determine the sustainability of the use of primary health services by the community. Patients who rate the quality of care as poor tend to have a lower interest in returning to take advantage of the same services in the future, and potentially switch to another health facility even if they have to travel longer distances. The long-term implications of this condition can have an impact on reducing the coverage of primary health services in a puskesmas working area if not immediately followed up. Therefore, the high proportion of unfavorable assessment in this study needs to be a serious concern for the manager of Ambacang Padang Health Center. (Ningsih, Hasmah, & Bhebhe, 2021)

If observed further, there is a similarity pattern between the frequency distribution of health cadre involvement and the quality of Health Services, where the proportion of low/poor categories in both variables is relatively balanced, each in the range of 54-57%. The similarity of this distribution pattern descriptively provides an initial idea that the two variables may have a relationship that needs to be analyzed further through bivariate statistical tests. From the perspective of Health System theory, the alignment of these patterns can be understood as an indication that the supporting elements of community-based services, such as health cadres, also form the foundation for the quality of Service perceived by patients at the primary health facility level. The conformity of these proportions reinforces the importance of in-depth study of the contribution of cadres to the quality of Public Health Services. (Nurdin, 2020)

Determination of low and high categories on the variables of cadre involvement, as well as the category of poor and good on the variables of service quality, in this study using the value of the cut off point based on the mean or median value of the total questionnaire score, as commonly used in cross-sectional quantitative research in the field of Public Health. This categorization method allows researchers to obtain a picture of the proportional distribution and can be compared with the results of similar studies at other locations. However, the use of data-based cut off point (mean / median) has limitations because the limit value of the category may vary depending on the distribution of data in each population of the study, so the proportion comparison between research needs to be done carefully. Understanding of these methodological limitations is important so that the interpretation of univariate results is not overgeneralized outside the context of the population of this study. (Erawati, 2024)

The univariate description in this study also reinforces the urgency of strengthening the capacity of health cadres as one of the strategies for improving the quality of primary health care, as found in a qualitative study on the experience of health cadres in detecting and preventing priority health problems at the community level. The study found that cadres who have high involvement tend to be more confident and more skilled in carrying out the function of early detection and health education to the community. Conversely, cadres with low involvement often feel that they lack adequate recognition and support from the formal health system, so they tend to carry out their duties passively or potluck. The findings provide additional context that

strengthens the understanding of the univariate picture of cadre involvement in this study. (Sukmawati, Hermayanti, Fadlyana, Maulana, & Mediani, 2025)

Researchers assume that the high proportion of low cadre involvement category in this study is most likely influenced by the characteristics of the Ambacang Padang Health Center working area located in urban areas with high population mobility, so that the interaction between cadres and residents becomes more limited than rural areas that tend to have closer social ties. The researchers also suspect that patients' assessment of poor quality of care does not fully reflect the technical performance of health workers, but is also influenced by higher patient expectations as access to health information through digital media increases today. In addition, researchers suspect that limited service time due to the high volume of patient visits also contributes to poor assessment of the dimensions of responsiveness and empathy of officers. These assumptions require further exploration through a qualitative approach to obtain a more comprehensive understanding of the root problems underlying the two univariate images.

The implication of this univariate picture shows that the Ambacang Padang Health Center needs to immediately design an intervention program that targets two sides at once, namely strengthening the capacity and motivation of health cadres through regular training and providing appropriate incentives, and improving the service system that focuses on increasing the dimension of responsiveness and empathy of health workers. The refreshing training Program for cadres needs to be scheduled regularly, accompanied by a structured mechanism for monitoring and evaluating cadre performance so that cadre involvement can be maintained and improved on an ongoing basis. On the service side, puskesmas need to evaluate the flow of services and the ratio of health workers to the number of patient visits, in order to ensure more efficient service time and better quality interaction with patients. With these measures, it is expected that the proportion of low categories in both variables can gradually decrease, resulting in an impact on improving the overall quality of public health services in the Ambacang Padang Health Center working area.

2. The Relationship Between the Involvement of Health Cadres With the Quality of Health Care According to Patient Assessment

The results of bivariate analysis in Table 3 showed that out of 52 respondents who assessed the involvement of low health cadres, as many as 38 people (73.1%) assessed the quality of Health Services was not good, while out of 44 respondents with high cadre involvement assessment, only 17 people (38.6%) assessed the quality of services was not good. Chi-Square test results obtained a value of $p=0.000$ ($p<0.05$), which shows that there is a statistically significant relationship between the involvement of health cadres with the quality of Health Services according to the assessment of patients in the work area of the Ambacang Padang Health Center. The value of the Odds Ratio of 4.31 (95% CI: 1.89–9.84) shows that patients who assess the involvement of low cadres have a 4.31 times greater risk of assessing the quality of health care is not good than patients who assess the involvement of high cadres. Confidence interval ranges that do not cross the number 1 (1.89–9.84) reinforce that the relationship is statistically valid and not caused by mere chance factors.

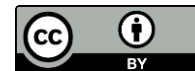


The meaningful relationship between the involvement of health cadres with the quality of health services in this study can be explained through the theory of service quality dimensions proposed by Parasuraman, where the ideal service quality includes five main dimensions, namely reliability, responsiveness, assurance, empathy, and physical evidence. Actively involved health cadres play an important role in strengthening the dimensions of empathy and responsiveness, because cadres become the closest socially and emotionally to the community before they access formal services at the puskesmas. When these dimensions are reinforced by the involvement of cadres, patients feel better understood and better prepared to face the service process, so that their assessment of the overall quality of service becomes more positive. Conversely, the lack of cadre involvement causes these dimensions to not be optimally met, which ultimately contributes to poor patient assessment of the quality of Health Services. (Nurdin, 2020)

The findings of this meaningful relationship are also in line with the results of research on the relationship of service quality with the interest in reusing health services, which found that non-clinical service support elements such as third-party involvement at the community level also significantly influenced patient assessment of the quality of primary health care. When the supporting elements are running optimally, the gap between the expectations and the reality of the service that the patient feels can be minimized, so that the assessment of the quality of service tends to be better. The similarity of the relationship patterns found in the study with the results of this study reinforces the external validity of the findings that the involvement of community-based support elements contributes significantly to the quality of health care perceived by patients. (Ningsih, Hasmah, & Bhebhe, 2021)

The value of the Odds Ratio of 4.31 obtained in this study shows the strength of the relationship is quite large, and in line with the pattern of findings in research on the relationship of Health Care Quality with patient satisfaction, which found that respondents with poor service quality assessment has several times greater risk of feeling dissatisfied than respondents who assess the quality of good service. The amount of risk that is consistent in the range of several times this confirms that the elements of service support, including the involvement of cadres, is not a variable with marginal influence, but a variable that is quite determinant in shaping the assessment of the quality of health services as a whole. The consistency of the amount of OR between these studies provides strong empirical evidence about the importance of intervention in the variable of cadre involvement as one of the strategies for improving service quality. (Sari, Pradiptha, & Triana, 2022)

The mechanism of the relationship between cadre involvement and service quality can also be explained through the role of cadres as an information bridge between the community and formal health facilities. Cadres who are actively involved in helping prepare patients for information before they access puskesmas services, for example through education about registration procedures, types of services available, and service schedules, so that the process of patient interaction with health workers becomes more efficient and targeted. Patients who have received assistance from cadres tend to have more realistic expectations of the services they will



receive, so the risk of gaps between expectations and the reality of services can be minimized. It is this mechanism that empirically explains why the involvement of cadres is significantly related to the assessment of the quality of services received by patients in the work area of the health center. (Nurmawati & Pramesti, 2022)

In addition, optimal cadre involvement contributes to the efficiency of the workload of health professionals in puskesmas, because some promotive and preventive functions, such as early detection and early education, have been carried out by cadres at the community level. When the workload of health workers can be distributed properly through the support of cadres, the time and attention given to each patient becomes more optimal, which has a positive impact on the patient's assessment of the dimensions of responsiveness and reliability of Service. Conversely, in areas with low cadre involvement, the entire burden of promotional and preventive services tends to be centered on health professionals, which risks reducing the quality of interaction between staff and patients due to time constraints. This condition also strengthens the meaningful relationship between low cadre involvement and poor service quality assessment in this study. (Khusnia & Nur, 2024)

The results of this study are also relevant to the findings of research in the working area of the Padang City Water children's Health Center, which shows that optimal cadre performance contributes to the successful implementation of community-based health programs, including community assessments of the effectiveness of services provided. The similarity of regional contexts, which are both located in the city of Padang, makes the findings as a geographically and culturally relevant comparison to the results of research at the Ambacang Padang Health Center. The consistency of the pattern of the relationship between the involvement/performance of cadres and the achievement of health programs in two different puskesmas working areas in the same city reinforces the generalization of the finding that the role of cadres is an important determinant for the success of primary health services in Padang City as a whole. (Raniwati, Ernawati, Sari, Sari, & Astuti, 2022)

From the perspective of community participation, the meaningful relationship between the involvement of cadres and the quality of health services in this study also strengthens the theory of community empowerment in the field of Health, which states that the success of community-based health services depends on the extent to which the community, including cadres, are actively involved in the entire service process. A qualitative study on the experience of health cadres in detecting and preventing priority health problems found that actively involved cadres were able to build public confidence in the formal health system, which in turn increased public acceptance of the quality of services provided. The trust that is built through the role of this cadre becomes one of the non-technical elements that help shape the patient's perception of the overall quality of Health Services. This confirms that the relationship between cadre involvement and service quality is not merely a statistical relationship, but has a strong conceptual foundation in the theory of Community Empowerment. (Sukmawati, Hermayanti, Fadlyana, Maulana, & Mediani, 2025)

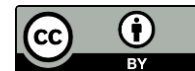


The meaningful relationship found in this study is also in line with the concept of continuum of care in the primary health system, where the continuity of service from the community level to formal health facilities largely determines the quality of the service experience felt by patients as a whole. Health cadres act as connecting nodes in the chain of service continuity, so that when these nodes are weak, service continuity becomes interrupted and has an impact on poor patient assessment of service quality in general. Conversely, when cadres are able to perform their function as a liaison optimally, patients feel a more complete and integrated service experience from the community level to health facilities. This concept reinforces the understanding that the quality of primary health care cannot be separated from the quality of involvement of supporting elements at the community level, including health cadres. (Novitasari & Rohmani, 2024)

The strength of the relationship shown by the OR value of 4.31 in this study also implies that interventions targeting the improvement of one variable, namely the involvement of cadres, have the potential to have a considerable impact on improving the overall quality of Health Services. This is consistent with the principle of public health interventions based on the key determinant approach, where the identification and intervention of variables with the highest relationship strength is considered more efficient than interventions that target many variables with smaller effects simultaneously. Thus, strengthening the involvement of cadres can be a top priority in efforts to improve the quality of health services in the Ambacang Padang Health Center working area, given the large contribution of this variable to the assessment of service quality based on the results of bivariate analysis. Prioritization of evidence-based interventions such as this is expected to result in improvements in service quality that are more measurable and efficient in terms of available resources. (Maameah, Rumayar, & Mandagi, 2022)

The researchers assume that the strong relationship between the involvement of health cadres and the quality of health services in this study is most likely due to the dual role of cadres as information providers as well as emotional liaisons between the community and health professionals, so that when the role is not running optimally, the impact is immediately felt on the entire range of service experiences received by patients. The researchers also suspected that patients who assessed low cadre involvement were most likely patients who came to the puskesmas without assistance or referrals from previous cadres, so they faced the service process without adequate initial information and tended to assess services more critically. In addition, the researchers suspect that the trust factor (trust) built by cadres to the community plays a role as a mediator in the relationship between cadre involvement and service quality assessment, although this mediator variable is not measured specifically in this study. These assumptions need to be studied further through research with a more complex design, for example by involving path analysis (path analysis) or a mixed method approach to understand the mechanism of these relationships in more depth.

The implication of this bivariate analysis results confirms that strengthening the involvement of health cadres needs to be one of the priority programs by the Ambacang Padang



Health Center in an effort to improve the quality of public health services in its work area. Operational steps that can be taken include the preparation of training and refresher programs for cadres on a scheduled basis, the provision of incentives and forms of proper appreciation, as well as the establishment of a system of supervision and continuous assistance between puskesmas officers and cadres in each assisted posyandu. In addition, puskesmas needs to build a feedback mechanism (feedback system) that involves cadres in routine evaluation of services, so that cadres not only act as program implementers, but also as strategic partners in making decisions related to improving service quality. With the targeted and continuous strengthening of cadre involvement, it is expected that the quality of public health services in the Ambacang Padang Health Center working area can be significantly and sustainably improved in the future.

CONCLUSIONS

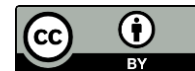
Based on the research results that have been described, it can be concluded that the involvement of health cadres in the Ambacang Padang Puskesmas working area is still mostly in the low category, with the proportion reaching 54.2% of the total respondents. In line with these findings, 57.3% of patients also assessed that the quality of health care they received was still poor, indicating that the quality of primary health care in the region did not fully meet community expectations. The results of the bivariate analysis reinforced this picture by showing a statistically significant relationship between health cadre involvement and health care quality according to patient assessment ($p=0.000$; $OR=3.98$; 95% CI: 1.76–8.99), which means that patients with low cadre involvement assessment have almost four times greater risk of assessing health care quality as poor than patients who assess high cadre involvement. Thus, it can be concluded that the involvement of health cadres has a significant effect on the quality of Public Health Services according to the perception of patients in the Ambacang Padang Health Center working area, so that strengthening the role of cadres is one of the strategic keys in efforts to improve the quality of Primary Health Services. Therefore, the Ambacang Padang Health Center needs to improve training, coaching, and monitoring the performance of cadres regularly, accompanied by the support of incentive systems and continuous supervision, in order to support the improvement of the quality of primary health services as a whole and continuously.

In practical terms, the results of this study provide benefits that can be directly applied by various related parties. For Puskesmas Ambacang Padang, these findings can be used as evidence-based basis in formulating internal policies related to cadre development programs, such as scheduling regular refresher training, preparing standard operating procedures (SOP) for cadre assistance, and allocating budgets for incentives and sustainable supervision systems. For the Padang City Health Office, the results of this study can be an input in formulating a more integrated health cadre development policy at the city level, given the pattern of cadre involvement problems that are similar to those found in other health centers in the city of Padang.



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